

# Ruixuan Zhang

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## EMPLOYMENT

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| <b>University of Utah, Department of Mathematics</b><br>Postdoctoral Research Associate, C.R. Wiley Instructor | July 2024–Present |
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## EDUCATION

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| <b>University of Kansas</b><br>Ph.D., Department of Mathematics<br>Dissertation: Conditional Limit Theorems for the KPZ Fixed Point with a Large Height.<br>Advisor: Zhipeng Liu | Aug. 2019–May 2024  |
| <b>Indiana University Indianapolis</b><br>Ph.D. Student in Statistics, Department of Mathematical Sciences<br>Supervisor: Pavel Bleher                                           | Aug. 2017–May 2019  |
| <b>Ningbo University</b><br>Bachelor of Management, Accounting, ACCA Class, School of Business<br>Bachelor of Science, Mathematics, School of Sciences                           | Sep. 2013–June 2017 |

## RESEARCH INTERESTS

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Kardar-Parisi-Zhang universality, integrable stochastic systems, random matrix theory, and probabilistic models from physics.

## PREPRINTS AND PUBLICATIONS

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- *Coalescence and fluctuations of O'Connell–Yor polymers via the free-energy correlation profile*, with Firas Rassoul-Agha, Xiao Shen, and Guangqu Zheng. [arXiv:2609.02064](https://arxiv.org/abs/2609.02064).
  - *Edgeworth-type expansion for the one-point distribution of the KPZ fixed point with a large height at a prior location*, with Ron Nissim. *Annales de l'Institut Henri Poincaré (B) Probabilités et Statistiques*, 62(2), 2026. [Journal](#); [arXiv](#).
  - *An upper tail field of the KPZ fixed point*, with Zhipeng Liu. *Communications in Mathematical Physics*, 406, 198 (2025). [Journal](#); [arXiv](#).
  - *Bayesian inference for thick-tailed stochastic volatility model—Empirical Study of Shanghai Composite Index*, with Wenli Huang. *Journal of Quantitative Economics*, 34(4).

## TEACHING EXPERIENCE

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| <b>University of Utah</b><br>2024–present      | MATH 3070, Applied Statistics I (Fall 2025, Fall 2026); MATH 1070, Introduction to Statistical Inference (Spring, Summer 2026); MATH 1030, Introduction to Quantitative Reasoning (Spring 2026); MATH 5090, Statistical Inference II (Spring 2025); MATH 3080, Applied Statistics II (Spring 2025); MATH 5080, Statistical Inference I (Fall 2024). |
| <b>University of Kansas</b><br>2019–2024       | Instructor: MATH 101, Algebra I (Fall 2019, Spring 2020); MATH 104, Pre-Calculus (Fall 2022); MATH 115, Calculus I (Fall 2020, Fall 2021); MATH 116, Calculus II (Spring 2021). Lab Instructor: MATH 126 (Spring 2023); MATH 127, Calculus III (Spring 2022, Spring 2024).                                                                          |
| <b>Indiana University Indianapolis</b><br>2018 | Instructor: MATH 165, Analytic Geometry and Calculus I (Fall 2018). Teaching Assistant: STAT 417, Statistical Theory, and STAT 350, Introduction to Statistics (Spring 2018).                                                                                                                                                                       |

## RESEARCH TALKS

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| 2026/03 | Seminar on Stochastic Processes, Union College — <i>The coalescence and exit time of O’Connell–Yor polymers via Gaussian integration by parts.</i>                       |
| 2025/09 | Stochastics/Discrete Analysis, North Carolina State University — <i>An upper tail field of the KPZ fixed point.</i>                                                      |
| 2025/09 | Statistics Seminar, Indiana University Indianapolis — <i>An upper tail field of the KPZ fixed point.</i>                                                                 |
| 2025/06 | Workshop on Stochastic Interacting Particle Systems and Random Matrices, Rényi Institute, Budapest.                                                                      |
| 2025/03 | Probability Seminar, Northwestern University — <i>An upper tail field of the KPZ fixed point.</i>                                                                        |
| 2024/02 | Probability Seminar, University of Utah — <i>Conditional limit theorem of the KPZ fixed point with a large height.</i>                                                   |
| 2023/11 | 22nd Northeast Probability Seminar, New York University — <i>A conditional limit theorem of the KPZ fixed point.</i>                                                     |
| 2023/10 | Probability Seminar, Texas Tech University — <i>Some exact formulas of the KPZ fixed point and its application.</i>                                                      |
| 2023/03 | Probability Seminar, University of Rochester — <i>Conditional limits of the KPZ fixed point with a large height at one location.</i>                                     |
| 2022/12 | 3rd Kansas Math Graduate Student Conference — <i>Asymptotics of the one-point distribution of the KPZ fixed point conditioned on a large height at an earlier point.</i> |
| 2022/11 | 21st Northeast Probability Seminar, Columbia University — <i>Conditional limits of the KPZ fixed point with height tending to infinity at one location.</i>              |
| 2022/04 | 2nd Kansas Math Graduate Student Conference, Kansas State University — <i>Right tail of the two-time distribution of the KPZ fixed point.</i>                            |
| 2021/12 | Frontier Probability Days, University of Nevada — <i>Right tail of the two-time distribution of the KPZ fixed point.</i>                                                 |

## HONORS AND AWARDS

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| <b>Outstanding Postdoc Award</b><br>Department of Mathematics, University of Utah                           | 2025 |
| <b>Paul F. Conrad Graduate Scholarship</b><br>Department of Mathematics, University of Kansas               | 2023 |
| <b>Graduate Student Summer Scholarships</b><br>Department of Mathematics, University of Kansas              | 2023 |
| <b>Graduate Student Travel Scholarships</b><br>Department of Mathematics, University of Kansas              | 2022 |
| <b>Outstanding Beginning Graduate Student Awards</b><br>School of Sciences, Indiana University Indianapolis | 2018 |
| <b>Xinghao Jiang Research Scholarships</b><br>School of Business, Ningbo University                         | 2015 |